

ARSHAD

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PROFESSIONAL SUMMARY

Final-year Backend Development student (TISS SSE) seeking an SDE / Backend Engineering internship. Built and deployed three production systems end-to-end: a Spring Boot/WebSocket messaging application (MongoDB, Docker, CI/CD on Render), a PHP/MariaDB Hospital Management System (deployed on InfinityFree), and a full-stack developer portfolio with a Spring Boot/MongoDB Atlas backend (deployed on Render and Netlify). Comfortable across REST API design, event-driven backend logic, and both relational and NoSQL databases.

EDUCATION

Bachelor of Software Development | Tata Institute of Social Sciences (TISS SSE) Expected 2027

Mumbai, India | Current GPA: 8.4/10 | 3rd Year

- Relevant coursework: Data Structures & Algorithms, OOP, Database Management Systems, Software Engineering.

KEY ACHIEVEMENTS

- **Production deployment:** independently built, containerized, and deployed a real-time messaging application to production (Spring Boot, WebSockets, MongoDB, Docker, Render CI/CD).
- **Security fix:** identified and closed a WebSocket identity-spoofing vulnerability by binding session authentication at the connection handshake, and independently root-caused a TOCTOU race condition in concurrent message handling.
- **Legacy system recovery:** diagnosed and fixed a privilege-escalation vulnerability, broken JOIN queries, and a failing bcrypt password-hashing flow in a live PHP/MariaDB system, then deployed it publicly on InfinityFree.

TECHNICAL SKILLS

- **Languages:** Java, PHP, JavaScript, C, HTML5, CSS3
- **Frameworks:** Spring Boot, Spring Security, Spring MVC, RESTful API Design, Native WebSockets
- **Databases:** MongoDB (NoSQL, Atlas), MySQL, MariaDB (Relational), single-collection schema design
- **Architecture:** Event-Driven Architecture (ApplicationEventPublisher), Concurrent Programming, OOP, Design Patterns (Repository, Factory, Singleton)
- **Reliability & Security:** Bean Validation, CORS Configuration, IP-Based Rate Limiting, Global Exception Handling, Health Checks
- **Testing:** JUnit, integration testing, Postman, PowerShell-based API testing
- **Tools:** Git, GitHub, Docker, Maven, Gradle, Ubuntu Linux (CLI), XAMPP, VS Code, Render, Netlify
- **AI Tools:** Claude, ChatGPT, and Gemini for research and code assistance; OpenCode for agentic scaffolding on personal projects
- **GenAI:** RAG, Vector Embeddings, MongoDB Atlas Vector Search
- **Concepts:** Data Structures & Algorithms, System Design, CI/CD pipelines, Agile/Scrum methodology

PROJECTS

Real-Time Enterprise Messaging Engine (Deployed) – Shipping Updates

Java, Spring Boot, WebSockets, MongoDB, Docker — [Live Demo](#) | [GitHub](#)

- Architected a real-time messaging backend by migrating from STOMP to native WebSockets, reducing per-message protocol overhead and simplifying the connection lifecycle for standard text frames.
- Built a message routing layer using a thread-safe ConcurrentHashMap for O(1) average-case lookup of active user connections.
- Implemented Spring Security authentication by binding JSESSIONID at the HTTP handshake level, preventing identity spoofing on WebSocket connections.
- Built a decoupled, event-driven architecture using Spring's ApplicationEventPublisher to move background processing (e.g. room cleanup, notifications) off the main request thread.

- Designed a unified MongoDB schema (single-collection design) to support both 1-on-1 direct messages and group chats with consistent query patterns.

Hospital Management System (Deployed)

PHP, MariaDB, HTML5, CSS3, JavaScript — [Live Demo](#) | [GitHub](#)

- Debugged and stabilized an MVC-restructured Hospital Management System, fixing routing errors, a privilege-escalation vulnerability, and broken JOIN queries across patient, doctor, and ward modules.
- Fixed a broken bcrypt password-hashing flow and rebuilt 22 missing view files by recovering and reconstructing logic from Git history, restoring core application functionality.
- Tested the full application locally using XAMPP, resolving remaining functional and UI bugs before deploying the final build publicly on InfinityFree.

Full-Stack Developer Portfolio (Deployed)

Java, Spring Boot, MongoDB Atlas, JavaScript, HTML5, CSS3 — [Live Demo](#) | [GitHub](#)

- Used OpenCode, an agentic AI coding tool, to scaffold a layered Spring Boot backend (Controller/Service/Repository/DTO/Exception), then independently root-caused and fixed MongoDB Atlas connection failure caused by a malformed URI and an unwhitelisted network-access IP.
- Resolved an ambiguous @ExceptionHandler conflict in Spring's global exception handling and implemented Bean Validation (@NotBlank, @Email, @Size), verifying correct HTTP responses across valid, invalid, and edge-case inputs.
- Traced a frontend-backend integration failure to a misrouted API call, then configured origin-restricted CORS (no wildcard) and IP-based rate limiting, validating both via direct API testing (PowerShell) and browser verification.
- Deployed the backend to Render (Java 21, Maven wrapper build) and the frontend to Netlify, replacing hardcoded localhost URLs with environment-based configuration for production.
- Diagnosed a 375px mobile CSS Grid overflow using Chrome DevTools, traced it to a source-order conflict between a base grid rule and a mobile media query, and shipped the fix through a Git feature-branch → PR → merge workflow.

WORK EXPERIENCE

Software Development Intern | PC Training Institute (PCTIL)

Aug 2025 – Oct 2025

- Built the billing and appointment-scheduling modules of a hospital management system backend in PHP and MySQL, working within an existing multi-module codebase.
- Optimized SQL queries and added indexing on frequently-filtered columns to speed up appointment and billing record lookups.
- Built a patient onboarding form with server-side validation to reduce manual data-entry errors during scheduling.

CERTIFICATIONS & CONTINUOUS LEARNING

RAG with MongoDB | MongoDB via Credly

August 2026

- Earned a verified MongoDB skill badge covering Retrieval-Augmented Generation (RAG), vector embeddings, semantic search, chunking, retrieval, and MongoDB Atlas Vector Search.

Software Development Job Simulation | Citi ICG Technology (via Forage)

April 2026

- Built a concurrent Java data-ingestion pipeline for streaming price updates, and a real-time JavaFX stock tracker integrated with the Yahoo Finance API using Google Guava caching.